Random Topology and Random Multiscale Mapping: An Automated Design of Multiscale and Lightweight Neural Network for Remote-Sensing Image Recognition

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***Abstract*— With the proposal of neural architecture search (NAS), automated network architecture design gradually becomes a new way in deep learning research. Due to its high capability regarding automated design, some pioneers have made an attempt to apply NAS in remote sensing and made some achievements, like 1-D/3-D Auto-convolutional neural network (CNN) and polarimetric synthetic aperture radar (PolSAR)-tailored Differentiable Architecture Search (PDAS). However, there are still some areas to be improved for existing NAS in remote-sensing field. In this article, we propose a random topology and random multiscale mapping (RTRMM) method to generate a multiscale and lightweight architecture for remote-sensing image recognition. First, a random topology generator generates the topology through random graph. Second, during the experiment, we find remote-sensing image features extracted by a multiscale network are more appropriate, compared with features extracted by a single-scale model. Nevertheless, the complexity inevitably increases with the introduction of a multiscale concept. Consequently, we design a variable search space consisting of decomposition convolution units under the guidance of mathematical analysis. The mapping of each neuron is then determined by a random multiscale mapping sampler. After that, we assemble the topology and mappings into blocks and construct three RTRMM models. Experiments on four scene classification datasets confirm the feature extraction capability and lightweight performance of RTRMM models. Moreover, we also observe that our approach achieves a better tradeoff between floating-point operations (FLOPs) and accuracy than some current well-behaved methods. Furthermore, the results on Vaihingen dataset verify the high feature-transfer capability.**

***Index Terms*— Deep learning (DL), multiscale and lightweight architecture, neural architecture search (NAS), random graph theory, remote sensing.**

Manuscript received May 13, 2021; revised June 30, 2021; accepted August 2, 2021. Date of publication August 13, 2021; date of current version January 31, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61725105 and Grant 41701508. *(Corresponding author: Kun Fu.)*

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Digital Object Identifier 10.1109/TGRS.2021.3102988

I. Introduction

BENEFITTING from the increase in computational capacities, deep learning (DL) technology ushers another revival, contributing to numerous remarkable achievements [1]–[11] in the field of computer vision (CV). Due to beneficial characteristics such as weight sharing and sparse connection, convolutional neural networks (CNNs) typically show excellent performance in regard to many image processing tasks. However, constructing a well-behaved deep CNN model is not a simple matter. Generally, remote-sensing images involve multiple data formats and modalities (e.g., hyperspectral imagery and synthetic aperture radar data) [12], [13], many special natures (e.g., rotation effect and rich background information) [14]–[18], and so on. The particularity of remote-sensing images needs to be taken into consideration during the design of neural networks [12]–[20]. Consequently, the construction process commonly requires strong expert knowledge and rich expert experience in remote sensing, making the procedure quite more challenging.

Recently, a novel automated framework for neural network architecture design called neural architecture search (NAS) [21]–[27] was proposed to relieve this pattern of complex and time-consuming architecture engineering. Generally, in the NAS framework, a search space is first defined. Some popular components of a deep neural network, such as convolution layer and pooling layer, are included in it. Then, a search algorithm needs to be carefully designed. It specifies the connection rules of a neural network and generates the final neural network architecture. Ultimately, the generated neural network is trained from scratch on a target dataset to test its

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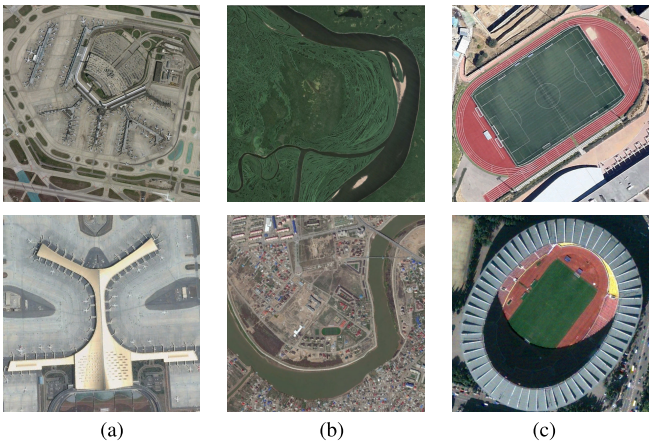


Fig. 1. Some fundamental challenges in remote-sensing scenes. (a) Widely covered scope. As shown in these two figures, they cover an airport, respectively, which can even reach millions of square meters. It is quite a large scope. (b) Background is diverse. The background of the two rivers is rather different, where one is characterized by vegetation and the other is characterized by urban areas, even though they are scenes of the same category. (c) Scene similarity. There exist many similar scenes in remote-sensing images, making it difficult to distinguish them for recognition algorithm. For example, both playground and stadium may consist of a playing field, but the major difference is that there are either grandstands surrounding the object or not. (a) Scope. (b) Background. (c) Similarity.

performance. Due to the characteristics in terms of a high level of automation, it has immediately attracted wide publicity and gradually become a research focus. An increasing number of novel and excellent NAS methods [23], [28]–[31], aiming to process natural scene image is rapidly emerging.

Different from natural scenes, there are substantial challenges for remote-sensing scenes. Under the influence of flying altitude and direction of the aerial vehicle, solar elevation angles, etc., the appearance of scenes may vary considerably. Imaging quality for remote-sensing image is greatly affected by the environment, comparing to natural scene image. Furthermore, remote-sensing images are often acquired in vertical view. This can cause the rotation effect of semantic contents reflected in them. While natural scene images are typically front view. Moreover, remote-sensing images usually cover a large complex scene. The background information is rich and varied, making the learner easily confuse different scenes. Some instances are detailedly illustrated in Fig. 1. Besides, scene similarity and relatively large image size also increase the difficulty of process for remote-sensing images.

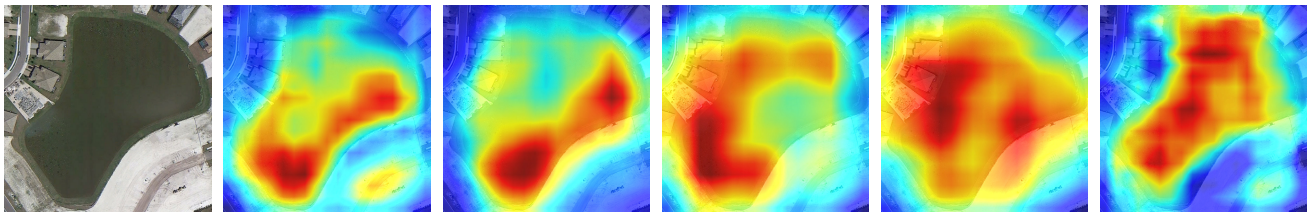
After experiencing the times of low-level [32], [33], mid-level [34], [35], and high-level [1]–[3] feature extraction methods, some pioneers have made an attempt to apply NAS theory in remote-sensing image-processing tasks to overcome the aforementioned challenges and improve the automation design level. So far many valuable works [36]–[39] have been done. These efforts greatly promote the development of remote-sensing image interpretation technology. However, there are still several areas to be improved. Currently, most of NAS methods in remote sensing even have strong human biases. The connection rules are strictly regulated and inflexible. Moreover, the scale information diversity is relatively deficient. As illustrated in Fig. 2, we find in the experiment that features captured by four types of single-scale networks do not match the *Pond* features expressed by the raw remote-sensing image whilst the multiscale model that contains these four scales almost extracts appropriate features of this *Pond*. It indicates that more consideration can be given to multiscale concept in NAS for remote-sensing images. Nevertheless, multiscale design inevitably makes the model more complex, leading to the generated neural network with a relatively high-computational complexity. This brings many quite intractable problems to the training stage. It is also adverse to the deployment of neural networks on some edge computing devices, like unmanned aerial vehicle (UAV).

In this article, we propose a random topology and random multiscale mapping (RTRMM) approach to generate a more lightweight neural network architecture with a capability of multiscale feature extraction. To be specific, we first utilize a random topology (RT) generator to generate an undirected graph depending on classical random graph models [40]–[43], which are widely used in daily life, such as Worldwide Web (WWW) and Preferential Attachment [44]. Different from graph convolutional networks (GCNs) represented by Hong *et al.* [45] which is a very novel and valuable work for hyperspectral images (HSIs), random graphs here are on spatial domain. Second, based on the experiment in Fig. 2, a random multiscale mapping (RMM) sampler is proposed, aiming to enrich the receptive field. Yet, the computational complexity inevitably increases even though the performance is improved. Thus, to alleviate this phenomenon, we design a variable multiscale search space consisting of decomposition convolution units. In addition, mathematical analysis is also given to support the decomposition design thought in theory. Ultimately, combining RT generator and RMM sampler, the entire neural network can be obtained. Series of experimental results suggest that the proposed RTRMM is a very competitive method. It outperforms baseline models which are designed by handcrafted processes or some NAS frameworks. And there is an obvious decline in terms of parameters quantity and floating-point operations (FLOPs) [46]. Our main contributions of this article can be concisely summarized as follows.

1) A novel RTRMM approach is proposed. It can automatically generate a more lightweight neural network architecture with a capability of multiscale feature extraction for remote-sensing image.
2) In the proposed approach, we first leverage the classical random graph theory to construct the network topological architecture. And guided by the experiment (as shown in Fig. 2), multiscale concept is also introduced to make the receptive field become more diverse.
3) To alleviate the growth of parameters quantity and FLOPs introduced by the multiscale design, a variable search space consisting of decomposition convolution units is defined. Theoretical analysis for the reason of decrease is given as well.

The rest of this article is organized as follows. Section II provides some related works about NAS in natural scene image and remote-sensing image. Section III introduces the proposed RTRMM method in detail. In Section IV, we conduct

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(a) (b) (c) (d) (e) (f)

Fig. 2. We carry out experiments on four types of single-scale models and a multiscale model which contains the foregoing four scales and list the corresponding heatmaps to analyze their feature extraction capabilities. The first column presents the raw remote-sensing image that contains a pond. The remaining columns successively show five different heatmaps obtained from five models, respectively. Here, SS is the abbreviation of single-scale and MS means multiscale. Best view in color. (a) Raw image. (b) SS1. (c) SS2. (d) SS3. (e) SS4. (f) MS.

experiments on five different remote-sensing image datasets to confirm the feature extraction and feature-transfer capability of the generated neural networks. Meanwhile, experimental settings and experimental analysis are detailedly depicted as well. Finally, conclusions are drawn in Section V and the plan for further work is also given in this section.

II. Related Work

In this section, some concise reviews of NAS in both natural scene image and remote-sensing image are given.

A. NAS in Natural Scene Image

With the proposal of NAS net (NASNet) [22], automated model design for natural scene image has gradually attracted more and more researchers' attention. Efficient neural architecture search (ENAS) [47] represents network architecture as a directed acyclic graph (DAG) and utilizes an efficient algorithm to make the search cost significantly reduced. Liu *et al.* [23] propose a simple but practical differentiable search method called Differentiable ARchiTecture Search (DARTS). This method draws support from bilevel algorithm [48], [49] to realize optimization. Due to the relatively little overhead and uncomplicated implementation, it promptly obtains wide acceptance and has a profound impact on the follow-up works, such as Progressive DARTS (PDARTS) [26], sequential greedy architecture search (SGAS) [27], and so on. Xie *et al.* [29] explore a novel model construction pattern with the help of random graphs under the guidance of some neuroscience studies [50], [51]. This approach achieves a very good performance and opens a different perspective to understand neural networks [52], [53]. Furthermore, it also brings inspirations in other aspects, like incremental learning [54], single image deraining [55], and so on. Apart from the above practice, some scholars leverage natural selection theory to establish NAS frameworks [25], [31], where models with well behavior are able to survive and multiply, otherwise they are eliminated. These methods reach a relatively high level as well.

B. NAS in Remote-Sensing Image

Inspired by the application of NAS in natural scene image, automated network design for remote-sensing image is also emerging. Chen *et al.* [36] elaborate 1-D/3-D Auto-CNN by leveraging the DARTS framework [23]. These two models are used as HSI classifier and achieve outstanding behavior on four hyperspectral datasets with a relatively small number of training samples. It is the first effort to explore an automated design procedure of CNNs for the HSI classification. Xie and Chen [37] first propose a gradient decent-based NAS strategy to search a neural network which specifically processes light detection and ranging (LiDAR) data. Under the guidance of this framework, the network design procedure facing LiDAR data becomes more automated and the searched model also obtains an obvious improvement on two widely-used datasets. Meanwhile, this approach expands the NAS method to cope with other remote-sensing data types as well. Besides LiDAR data, Dong *et al.* [38] introduce a polarimetric synthetic aperture radar (PolSAR)-tailored Differentiable Architecture Search (PDAS) method. It is also the first time to introduce the thought of architecture search into the field of PolSAR data processing. Experiments actively demonstrate that this approach has a very competitive performance on three widely-used PolSAR benchmarks.

Motivated by the achievements of random graphs and considering the corresponding limitations during the experiment, we introduce more abundant scale information and lightweight design thought in the architecture construction procedure. The mathematical analysis provides a theoretical explanation for our RTRMM approach. And the experimental results also confirm its superiority and effectiveness.

III. Methodology

The overview of our RTRMM method is shown in Fig. 3. RT generates the topological relation of neural network blocks in a random way and RMM determines the mappings of all neurons in it. Finally, these blocks can be stacked to construct the entire architecture. These contents will be described in detail below.

A. RT Generator

The RT generator is on the basis of random graph theory, which is a different perspective to construct a neural network topology for remote-sensing image. Actually, the RT generator can be viewed as a mapping φ from the random graph parameter space Θ and a set of learnable architecture parameters $\mathcal{M}$ to a space of neural network architectures $\mathcal{A}$ under the condition of a pseudorandom seed s. When a $\theta \in \Theta$, a $\boldsymbol{\mu} \in \mathcal{M}$ and a value for the parameter s are given, $\varphi(\theta, \boldsymbol{\mu}|s)$ returns

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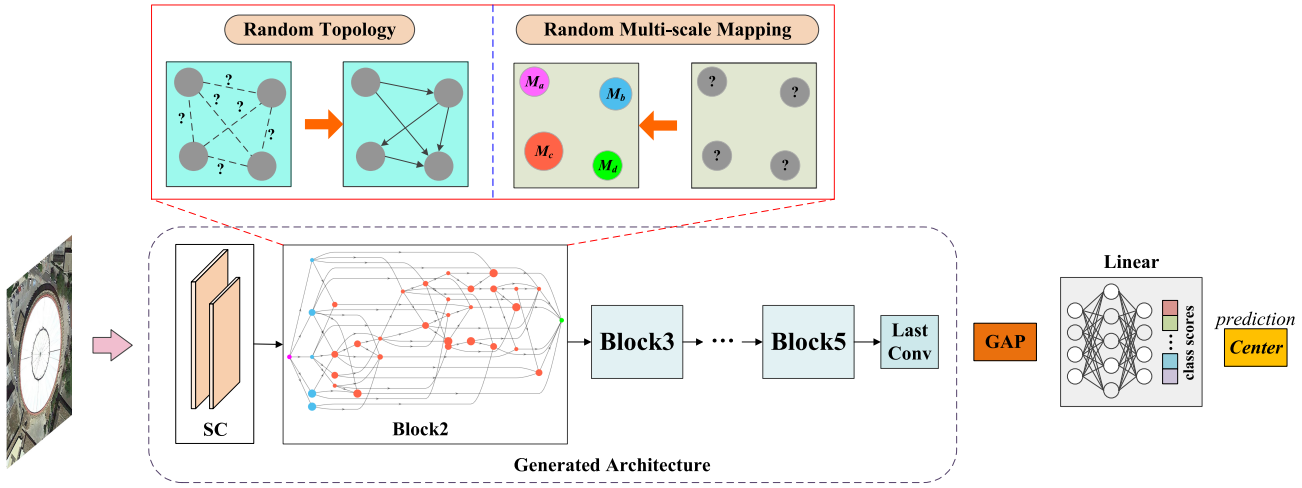


Fig. 3. Overview of the proposed RTRMM method. It shows the procedure of the RTRMM module. The topological relation is more flexible and the scale is more diverse. Besides, we also introduce a lightweight concept into this method. Here, **Block1** is a stem convolution (SC) layer. In **Block2** of the generated architecture, circles of different sizes mean different scales. And different colors indicate different types of nodes. Note that, the generation process of **Block3** to **Block5** are similar to **Block2**. For convenience, we just show the details for **Block2**. Best view in color.

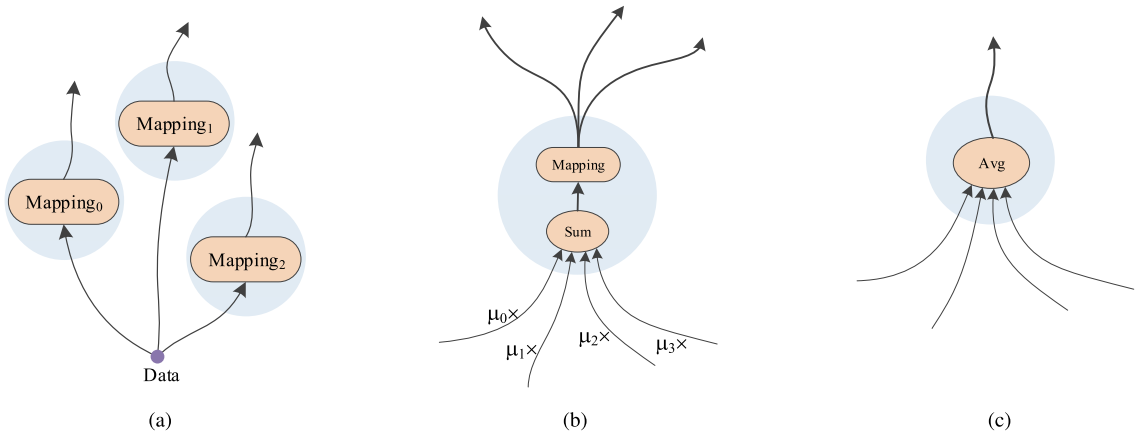


Fig. 4. Inner structures of input node, middle node, and output node are shown from left to right. (a) Input node. (b) Middle node. (c) Output node.

a concrete neural network architecture $a \in \mathcal{A}$. Notice that, at this moment, there are only topological relationships in this architecture a; mappings, utilized to transform features, like convolutions or poolings, of this architecture are ignored temporarily. The set $\mathcal{A}$ is typically a group of similar networks. This can be interpreted as a black box process. θ, s and $\boldsymbol{\mu}$ are the input arguments while a is the output result. Different from some reinforcement learning (RL) search methods, the topological structure is determined by the machine itself. To reduce the artificial biases in the procedure of architecture generation, we do not compulsively stipulate the indegrees or outdegrees of the graph in advance. They are just subject to a probability distribution. This allows for a higher degree of freedom of the connection relationships. Moreover, it is also quite beneficial to the diversity of architecture.

In general, a neural network can be regarded as a DAG from the perspective of gradient propagation. However, the architecture generated by the RT generator is an undirected graph. Therefore, to convert it to a DAG, we utilize a simple and natural rule: we label the nodes in the undirected graph with a fixed sequence number starting at 0 and stipulate that the direction of every edge points from a small index to a large index. This definition guarantees not to form closed loops in the generated neural networks. Hence, the forward propagation can flow from input node to output node without obstacle. Meanwhile, the backpropagation [56] can be carried out normally as well.

In an architecture, nodes are designed as shown in Fig. 4. There are three types of nodes in our approach. It is worth mentioning that there may be more than one input nodes, as shown in Fig. 4(a). For the data sent into input nodes, we do not change it and merely make it flow normally. The output of an input node y_l can be computed by

$$y_l = \mathbb{M}_l(x), \quad l = 1, 2, \ldots, L \tag{1}$$

where x is the input data, L is the number of input nodes, and $\mathbb{M}_l$ is the mapping of an input node.

An example of a middle node is given in Fig. 4(b). Here, μ_0, μ_1, μ_2, and μ_3 which indicate the importance of these four input edges are learnable architecture parameters. From the perspective of graph theory, they represent the parameters of edges. They reflect which input branches are more important

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to the middle node. Accordingly, more attention will be paid to those relatively important input features. Then, we apply a transfer function for these parameters and they are multiplied by the corresponding input data x_i (The input of each edge is different.) on each edge separately and added together. After that, the mapping of this middle node transforms the weighted sum obtained by x_i and μ_i. The mapping selection process will be discussed in Section III-B. Ultimately, we make n copies [$n = 3$ in the example in Fig. 4(b)] of this result and send them out to the next nodes. We emphasize that the practical indegree and outdegree of each node may not be identical with this instance because of randomness. This illustration is just convenient for us to explain the inner structure of the middle node. Besides, with the aspect of parameters update, the learnable architecture parameters are optimized as general weights and biases in a neural network in the backpropagation stage by data themselves. Abstractly, we have

$$y_j = \mathbb{M}\left(\sum_i \sigma(\mu_i)\cdot x_i\right), \quad i = 1, 2, \ldots, I, \quad j = 1, \ldots, J \tag{2}$$

where σ represents the transfer function, I is the number of input branches, J is the number of duplicates, and $\mathbb{M}$ is the mapping of this node.

For the output node, we collect all nodes whose outdegree is 0 and average the output of these nodes as the final output result [see Fig. 4(c)]. This can be written as

$$y = \frac{1}{Q}\cdot \text{sum}(x_1, x_2, \ldots, x_Q) \tag{3}$$

where Q is the number of nodes with the outdegree of 0.

B. RMM Sampler

After the architecture topology is established, specific mappings are determined by the RMM sampler. From the experimental results, e.g., as shown in Fig. 2, we find that the diversity of scale is conducive to feature extraction for remote-sensing image owing to its complex and unique characteristics, such as rich background information, high scene similarity, and so on. We define that the search space consists of several different scales of depthwise (DW) convolutions [57], followed by a pointwise (PW) convolution [57]. In DW convolutions, the convolution operation is only applied in the corresponding channels between the input feature map and convolution filter. As a result, the number of output channels is equal to the number of input channels. DW convolutions are powerless to change the number of channels, even though they own many good properties (few parameters, high efficiency, etc.). Thus, a PW convolution is completely necessary. In fact, PW convolution is the same as a common convolution except for the kernel size which is one in width and height.

Nevertheless, the computational complexity inevitably increases with the addition of multiscale concept. Therefore, we adopt a lightweight design under the guidance of mathematical analysis to make the model more compact. Concretely, we decompose the large convolution kernel with the receptive field of it remaining the same. As exemplified in Fig. 5, a $K \times K$ convolution kernel can be divided into $(K-1)/2$ 3×3 convolution kernels, where $K \in \{2t+1, t \in \mathrm{N} \text{ and } t \geq 2\}$. (In general, the width and height of the convolution operation are equivalent.) Hence, parameters and FLOPs [46] of these two types of convolution can be calculated as

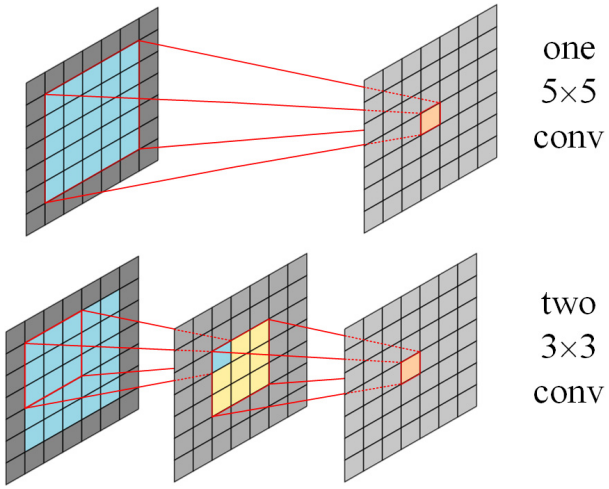


Fig. 5. Sketch of the 5×5 convolution kernel decomposition process. one 5×5 convolution kernel can be decomposed into two 3×3 convolution kernels. It can be seen that the receptive field is not changed before and after the decomposition. Best view in color.

$$\text{Parameters}(U) = C_{\text{in}} \cdot K^2 \tag{4}$$

$$\text{Parameters}(D) = C_{\text{in}} \cdot \frac{K-1}{2} \cdot 3^2 \tag{5}$$

$$\text{FLOPs}(U) = (2K^2 - 1) \cdot C_{\text{in}} \cdot H_{\text{out}} \cdot W_{\text{out}} \tag{6}$$

$$\text{FLOPs}(D) = \frac{K-1}{2} \cdot (2 \cdot 3^2 - 1) \cdot C_{\text{in}} \cdot H_{\text{out}} \cdot W_{\text{out}} \tag{7}$$

where U means undecomposed convolution operation and D indicates the convolution with decomposition. C_{in}, H_{out}, and W_{out}, respectively, represent the quantity of input channels, the height of output feature map, and the width of output feature map. It is noted that convolution is *SAME* mode where the size of features does not change after convolution, and bias is excluded just for simplicity here. From the above four formulas, regarding the quantity of parameters and FLOPs metric, we can find that the latter design pattern is more compact, that is

$$\text{Parameters}(U) - \text{Parameters}(D) > 0, \quad K > 3 \tag{8}$$

$$\text{FLOPs}(U) - \text{FLOPs}(D) > 0, \quad K > 3 \tag{9}$$

yet the receptive fields of these two cases are both $K \times K$ pixels.

Apart from DW and PW convolutions, rectified linear unit (ReLU) and batch normalization (BN) [58] are also attached in a specific mapping. For the sake of simplicity, we use a small unit called decomposition convolution unit to denote a specific multiscale mapping in the search space. Overall, the search space and a small unit are illustrated in Fig. 6. Regarding the selection of the mapping for each node, we adopt a random sampling with replacement method. What's more, to ensure the fairness and objectiveness of selection, we set the probability

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TABLE I

ENTIRE NETWORK CONFIGURATION. SC IS THE ABBREVIATION OF STEM CONVOLUTION. COLUMN "BLOCK" SHOWS THE INTERNAL STRUCTURE OF EACH **BLOCK**. "CONV" MEANS COMMON CONVOLUTION OPERATION. "NODES" MEANS THE NUMBER OF NODES IN A RTRMM RANDOM GRAPH. H AND W DENOTE HEIGHT AND WIDTH OF THE RAW INPUT REMOTE-SENSING IMAGE, RESPECTIVELY. C AND C' ARE HYPERPARAMETERS. "SEARCH SPACE" LISTS THE AVAILABLE CONVOLUTION SCALES OF THE CORRESPONDING **BLOCKS**

Name	Input shape	Block	Nodes	Output channels	Search space
SC (Block1)	$H \times W$	3×3 Conv-BN	-	$\frac{C}{2}$	-
Block2	$\frac{H}{2} \times \frac{W}{2}$	RTRMM random graph	N	C	$\{K_{max}, \cdots, 3\}$
Block3	$\frac{H}{4} \times \frac{W}{4}$	RTRMM random graph	N	C	$\{K_{max}-2, \cdots, 3\}$
Block4	$\frac{H}{8} \times \frac{W}{8}$	RTRMM random graph	N	$2C$	$\{K_{max}-4, \cdots, 3\}$
Block5	$\frac{H}{16} \times \frac{W}{16}$	RTRMM random graph	N	$4C$	$\{K_{max}-6, \cdots, 3\}$
Last Conv	$\frac{H}{32} \times \frac{W}{32}$	ReLU-PW-BN	-	C'	-
GAP	$\frac{H}{32} \times \frac{W}{32}$	Global Average Pool	-	C'	-
Linear	$1 \times C'$	Fully Connected	-	-	-

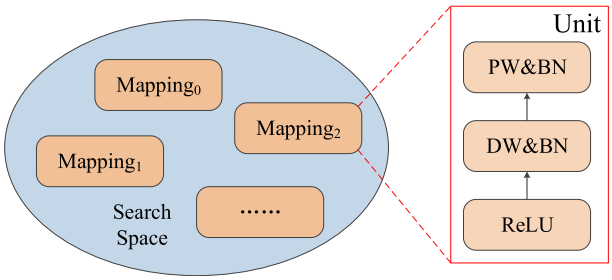


Fig. 6. Illustration of the search space and the composition of a small unit.

of each mapping in the search space equal to one another

$$\mathbb{M}_v = \text{SAMPLE}(V, 1), \quad v = 1, \ldots, V \tag{10}$$

where V stands for the quantity of mappings in the search space. SAMPLE$(V, 1)$ represents random sampling with replacement, but only selecting one element, i.e., one small unit from the search space.

C. Entire Architecture

Table I presents the entire architecture configuration in detail. Here, we define a RTRMM random graph as a block. The entire network is composed of several RTRMM blocks. Moreover, the maximum scale, namely $K_{\max}$, in the search space attached to each block decreases with the deepening of network layers for the size of feature maps is gradually decreasing. With regard to the loss function, we utilize cross entropy (CE) with label smoothing to minimize the margin between predicted results and the ground truth (GT) labels.

To summarize, we can utilize the following items to represent the entire architecture:

$$\text{architecture} := (\boldsymbol{\varphi}(\theta, \boldsymbol{\mu}|s), \mathbb{M}, \boldsymbol{\omega}) \tag{11}$$

where $\boldsymbol{\varphi}$ represents all RTRMM random graphs the network contains, $\mathbb{M}$ is a mapping set of all nodes, and $\boldsymbol{\omega}$ denotes the weights and biases of the entire network. The overall procedure is organized in Algorithm 1, where the generation phase aims to search the entire architecture and the optimization phase is to train the model by the classical backpropagation algorithm [56].

Algorithm 1 Overall Procedure of RTRMM

Require:
Random graphs parameters θ and pseudorandom seed s; the maximum scale K_{max}; the number of channels C and C'; training set and validation set

Ensure:
Trained entire architecture $(\boldsymbol{\varphi}(\theta, \boldsymbol{\mu}^*|s), \mathbb{M}, \boldsymbol{\omega}^*)$.

Generation Phase:
$\boldsymbol{\varphi}(\theta, \boldsymbol{\mu}|s) \leftarrow$ Random Topology generator;
$\mathbb{M} \leftarrow$ Random Multi-scale Mapping sampler;
obtain untrained entire architecture.

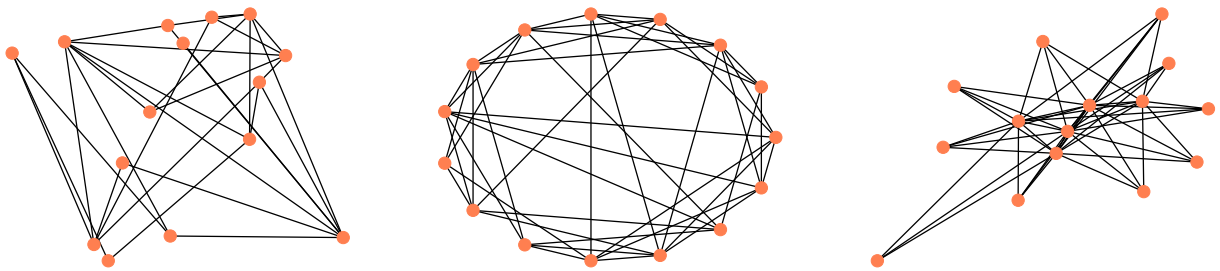
Optimization Phase:
initialize $\boldsymbol{\omega}$, $\boldsymbol{\mu}$, learning rate ξ, train epochs E;
for every epoch in E: // jointly optimized
    $\boldsymbol{\mu}, \boldsymbol{\omega} \leftarrow \boldsymbol{\mu}, \boldsymbol{\omega}$ - $\xi \nabla_{\boldsymbol{\mu}, \boldsymbol{\omega}} L_{train}(\boldsymbol{\mu}, \boldsymbol{\omega})$
select the best $\boldsymbol{\mu}^*$, $\boldsymbol{\omega}^*$ according to the performance on the validation set.

D. Overview of Random Graphs

In this section, we briefly review several popular random graphs to make the method description more comprehensive and make it easier to be grasped.

1) **Erdős–Rényi (ER)** random graph [41]. We assume that there are N nodes in an ER random graph. All edges

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(a) (b) (c)

Fig. 7. Three visualization examples: (a) ER random graph, (b) WS random graph, and (c) BA random graph. Variables of these random graphs are stated in the title of subfigures. It is noted that, these examples are just demos. The practical topology is more complicated than that. (a) ER $(15, 0.3|100)$. (b) WS $(15, 6, 0.3|100)$. (c) BA $(15, 10|100)$.

between every pair of nodes are subject to independent identical distribution (i.i.d.). And the connected probability of each node is P. So we can express an ER random graph as $\mathrm{ER}(N, P|s)$, where s is a pseudorandom seed.

2) **Watts–Strogatz (WS)** random graph [42]. WS random graph can be written as $\mathrm{WS}(N, R, P|s)$. The meaning of N and s are the same as mentioned above. R represents the number of neighbors of one node. At first, all nodes are arranged in a ring and each node is connected to its $R/2$ neighbors in clockwise and counterclockwise order, respectively. So R must be an even number. Next, for a node n_i, choose a node n_j in clockwise direction with probability P from all nodes that are unconnected to n_i. Then, the two nodes are connected and the connection of n_i and one of its neighbors is broken. This process is repeated $R/2$ times. After traversing all nodes, the WS random graph is generated.

3) **Barabási–Albert (BA)** random graph [43]. Also, we presume that the BA random graph contains N nodes. Initially, there are only T $(1 \leq T < N)$ nodes and all relationships do not exist at the moment. Then, a new node with T edges is added in this incomplete random graph. The connection between this node and other nodes is related to all existing nodes' degree. A relatively large degree indicates a relatively high connection probability and vice versa. The generation of a BA random graph is done until $N-T$ nodes have been added. Thus, we can use the expression $\mathrm{BA}(N, T|s)$ to denote a BA random graph.

Three visualization examples of these three random graphs are separately shown in Fig. 7.

IV. Experiments

In this section, we conduct several experiments on four extensively used remote-sensing scene classification datasets to confirm the feature extraction capability of RTRMM. And then, experiments on a challenging remote-sensing semantic segmentation dataset are performed to verify the feature-transfer capability of the proposed method. Quantitative results and the corresponding qualitative results are also shown. In addition, we present the explanations and analyses of these experiments in detail as well.

A. Datasets

We evaluate the proposed RTRMM method on UC Merced (UCM) [59], SIRI-WHU [35], [60], [61], aerial image dataset (AID) [62], NorthWestern Polytechnical University (NWPU)-RESISC45 (NWPU45) [63], and the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen dataset [64]. The introduction and settings in experiment for these datasets are listed below.

1) UCM: The UCM[1] dataset released in 2010 is a classical remote-sensing scene classification dataset, where 2100 remote-sensing images with a pixel resolution of 1 foot are collected from the United States Geological Survey (USGS) National Map. These images evenly disperse in 21 categories, that is 100 images for each category. The width and height of each image are both 256 pixels. Like the previous work [59], we randomly select 1680 remote-sensing images as training set and remaining images are used for evaluation.

2) SIRI-WHU: SIRI-WHU[2] is a 12-category dataset organized by the RS_IDEA Group in Wuhan University and it covers many urban regions in China. There are 200 images which are extracted from Google Earth for each category. Each image has a fixed size of 200×200 pixels and the spatial resolution is 2 m. We use 50% of the total SIRI-WHU samples for training and the other for test, according to the statements issued by the dataset publishers [35].

3) AID: AID[3] is a large-scale and a widely used remote-sensing image scene classification dataset. The size of each image is 600×600 pixels which is a fairly large image size and the spatial resolution changes from 8 to 0.5 m. The quantity of remote-sensing images in AID reaches 10 000. These images disperse in 30 categories and each category consists of 220 to 400 images. Smaller interclass dissimilarity, higher intraclass variations and unbalanced distribution of categories make AID highly challenging for current DL methods. Based on the document of AID [62], we use 50% of the images as training set and the remaining 50% as test set.

4) NWPU-RESISC45: NWPU45[4] is a quite large-scale dataset which is obtained from Google Earth, under the guidance of many experts in remote-sensing image interpretation.

[1] http://weegee.vision.ucmerced.edu/datasets/landuse.html
[2] http://www.lmars.whu.edu.cn/prof_web/zhongyanfei/Num/Google.html
[3] http://captain.whu.edu.cn/project/AID/
[4] http://pan.baidu.com/s/1mifR6tU

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Totally, there are 31 500 remote-sensing images that cover many countries and regions in the world. These images are divided equally into 45 different categories, i.e., 700 images for each category. The size of each remote-sensing image is 256×256 pixels and the spatial resolution ranges from about 30 to 0.2 m. Consistent with the original article [63], we just employ one fifth of all samples to train the generated network and the remaining four fifths are used for testing. Relatively few training samples, rich image variations and a series of other characteristics make this dataset more difficult to be processed.

5) ISPRS Vaihingen: The ISPRS Vaihingen[5] dataset consists of 33 large-scale remote-sensing images with a spatial resolution of 9 cm. The average width and height of each image are 2064 pixels and 2494 pixels, respectively. These images are obtained over the small village of Vaihingen in Germany. Each image has three types of bands which correspond to near infrared, red, and green wavelengths. In addition, digital surface models (DSMs) are also provided as complementary data. Notably, only 16 of these 33 images are provided with annotation information on a per-pixel basis. These annotations include six semantic categories: *Impervious Surfaces*, *Building*, *Low Vegetation*, *Tree*, *Car*, and *Clutter/Background*. Similar to the previous works [65]–[67], we divide the 16 labeled images into two parts. One which comprises 11 images with index in {3, 5, 7, 13, 17, 21, 23, 26, 32, 37} is used for training. The other, namely images with index in {11, 15, 28, 30, 34}, serves as the test set.

B. Evaluation Metrics

Like most of the related work on remote-sensing scene classification, we adopt the overall accuracy (OA) metric as well as the quantity of parameters and FLOPs to evaluate the generated model. OA is to judge a model's performance in classification, which can be computed by

$$\mathrm{OA} = \frac{N_{\mathrm{correct}}}{N_{\mathrm{all}}} \tag{12}$$

where N_{correct} is the number of correctly classified samples and N_{all} is the number of all samples. The latter two metrics are used to evaluate its behavior in terms of lightweight characteristics. Here, FLOPs for DW convolution can be referenced in (6). Besides, FLOPs for PW convolution (i.e., common convolution with a kernel size of 1×1) and fully connected (FC) layer can be computed separately as

$$\mathrm{FLOPs(PW)} = (2C_{\mathrm{in}} - 1) \cdot H_{\mathrm{out}} \cdot W_{\mathrm{out}} \cdot C_{\mathrm{out}} \tag{13}$$

$$\mathrm{FLOPs(FC)} = (2F_{\mathrm{in}} - 1)F_{\mathrm{out}} \tag{14}$$

where, the meaning of C_{in}, H_{out}, and W_{out} are identical to (6) and bias is not considered as well. C_{out}, F_{in}, and F_{out} are the number of output channels, input neurons, and output neurons, respectively. It is remarkable that nonlinearity mapping is computed for free.

Regarding the semantic segmentation task, we utilize three widespread employed metrics: mean F_1-score, mean Intersection-over-Union (mIoU), and OA. First, we let TP, FN, and FP separately stand for the number of correctly classified positive samples, the number of positive samples which are misclassified as negatives, and negative samples which are recognized as positives. Based on them, the value of Recall and Precision are formulated as

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{16}$$

Then, we can compute the F_1-score and IoU with the following formulas:

$$F_1 = (1 + \beta) \cdot \frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\beta^2 \cdot \mathrm{Precision} + \mathrm{Recall}}, \quad \beta = 1 \tag{17}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{18}$$

Ultimately, the mean F_1-score and mIoU can be obtained by averaging the F_1-score and IoU values over all categories. Typically, the higher mean F_1-score indicates the better segmentation performance. And similarly, a large mIoU also means a better result.

C. Implementation Details

We set $P = 0.2$ for the ER random graph, $R = 4$, $P = 0.75$ for the WS random graph and $T = 5$ for the BA random graph to be consistent with [29] which found that the effect of these hyperparameters is relatively better. It is noted that these hyperparameters of graphs do not participate in optimization. And N is 32 for all the three random graphs. The transfer function σ is set to *logistic*. From this, three topologies are generated. Next, we set $K_{\max} = 9$ in our experiments, which means that the largest search space consists of four scales: 3×3, 5×5, 7×7, and 9×9. This is in the light of [68] which found that 9 is a critical scale. Moreover, C and C' are adjusted to 32 and 128 separately. Therefore, three types of neural networks are obtained: **ER(32, 0.2)-4**, **WS(32, 4, 0.75)-4**, and **BA(32, 5)-4**. Fig. 8 shows the details about Block2s of three generated architectures with their mappings. For convenience sake, we next only show Block3, Block4, and Block5 of **BA(32, 5)-4** model in Fig. 9. And, as an analogy, the remaining blocks of **ER(32, 0.2)-4** model and **WS(32, 4, 0.75)-4** model can also be obtained. In the training stage, we just use kaiming normal [69] to initialize three RTRMM models' parameters, including learnable architecture parameters and weight parameters, while weights and biases of other high-level feature-based methods and NAS methods are pretrained on ImageNet [70]. For the optimizer, we choose momentum stochastic gradient descent (SGD) [71] with a cosine schedule [72] where the basic learning rate is 0.05. Meanwhile, to avoid over-fitting problems, cutout [73] is applied to the training procedure on UCM and SIRI-WHU. About the iterations, we train all these three generated networks for 200 epochs on four remote-sensing image scene classification datasets.

With regard to the semantic segmentation task, we replace the original backbone of fully convolutional network (FCN) [9] with the proposed three RTRMM models to carry

[5]http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

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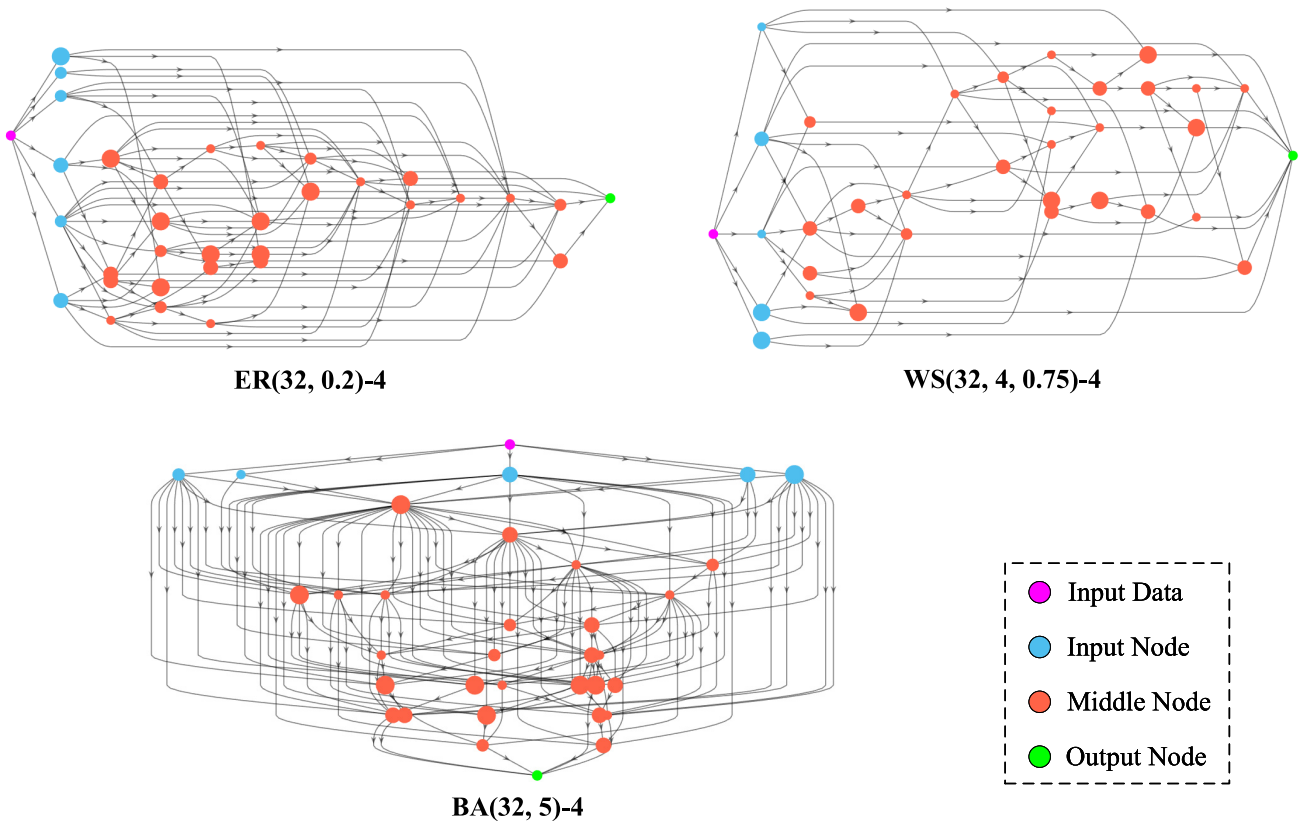


Fig. 8. Block2s of three generated architectures with their mappings. A larger node size represents a larger scale and vice versa. Best view in color.

out experiments on the ISPRS Vaihingen dataset. In the aspect of initialization, we load the ImageNet pretrained model on original FCN whilst we just load NWPU45 pretrained weights on FCNs with three RTRMM models as the backbone. For the experimental settings, following the prior achievements [74], [75], we employ the SGD optimizer and set the momentum to be 0.9 so that the gradients can be smoothed. Meanwhile, the weight decay is also set to 10^{-3} to prevent over-fitting phenomenon. Due to the GPU memory limitation, we randomly crop the size of input patches from the original images to patches of 513×513 pixels before they are sent to the segmentation networks. Then, all the compared models are trained for 200 epochs. During the training stage, the learning rate which is initialized to 0.01 is dropped with a poly-scheduler with the power of 0.9.

D. Scene Classification Experiments

Except the proposed approach, we also evaluate some popular low-level, mid-level, high-level and NAS methods. All the experimental results, including many evaluation metrics, on the UCM, SIRI-WHU, AID, and NWPU45 scene classification datasets are listed in Tables II–V, respectively. Here, **BA(32, 5)-4**, **ER(32, 0.2)-4**, and **WS(32, 4, 0.75)-4** are abbreviated as BA-4, ER-4, and WS-4, respectively.

From these results, we can see that the proposed three RTRMM models reach a higher OA performance on four datasets with the fewest parameters and FLOPs. To be specific, the WS-4 model shows great performance on UCM and SIRI-WHU while BA-4 surpasses various baseline models on AID and NWPU45. In all comparison methods, we can find that some methods also have a good behavior in remote-sensing field, but the proposed three RTRMM models achieve a better performance. For some recently proposed well-behaved NAS methods, such as DARTS, PDARTS, and SGAS, the gap of the performance, compared to the proposed RTRMM models, is about one percentage point or more. In these approaches, relatively speaking, the scale informa-

TABLE II
EXPERIMENTAL RESULTS ON THE UCM DATASET

Network	Level	OA(%)	Params(M)	FLOPs(G)
GIST [62]	Low	46.90±1.76	-	-
LDA (SIFT) [62]	Mid	69.52±1.27	-	-
IFK (CH) [62]	Mid	83.79±2.32	-	-
AlexNet [62]	High	95.02±0.81	40.31	0.91
VGGNet [62]	High	95.21±1.20	134.35	20.21
GoogLeNet [62]	High	94.31±0.89	5.62	1.98
ResNet [4]	High	95.90±0.87	23.55	5.38
DenseNet [5]	High	**96.76±0.49**	6.98	3.77
MobileNetV2 [76]	High	95.71±0.40	2.25	0.41
NASNet [22]	NAS	89.62±0.36	4.26	0.77
PDARTS [26]	NAS	91.52±0.96	4.19	0.73
SGAS [27]	NAS	92.05±0.89	4.69	0.81
ALP [39]	NAS	93.43±0.71	2.63	-
MNASNet [28]	NAS	94.52±1.29	3.13	0.43
DARTS [23]	NAS	95.19±0.53	3.97	0.71
RTRMM(BA-4)	NAS	96.33±0.89	0.82	0.39
RTRMM(ER-4)	NAS	96.43±0.84	**0.81**	**0.38**
RTRMM(WS-4)	NAS	**96.76±0.56**	0.82	**0.38**

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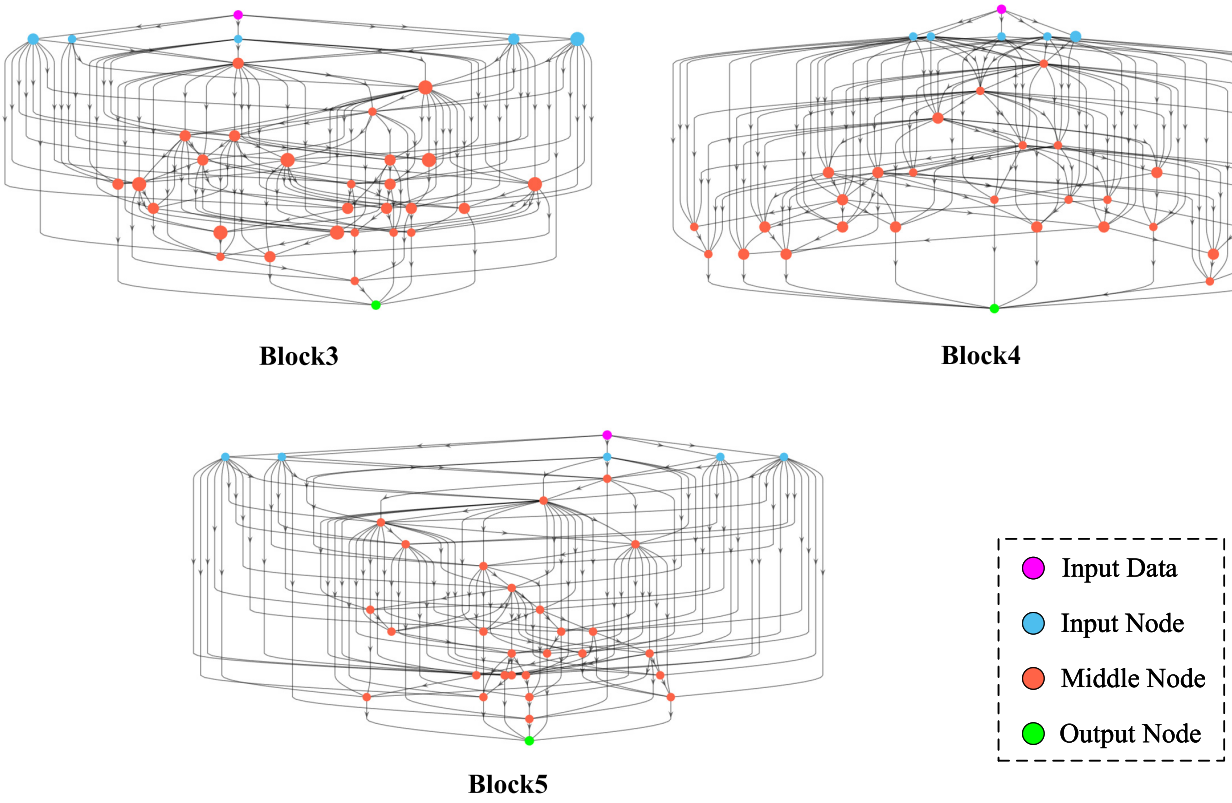


Fig. 9. Block3, Block4, and Block5 of **BA(32, 5)-4** model. Best view in color.

TABLE III
EXPERIMENTAL RESULTS ON THE SIRI-WHU DATASET

Network	Level	OA(%)	Params(M)	FLOPs(G)
AlexNet [1]	High	90.23±0.17	40.27	0.53
VGGNet [2]	High	90.70±0.60	134.31	12.29
GoogLeNet [3]	High	89.65±0.33	5.61	1.17
ResNet [4]	High	91.38±0.56	23.53	3.52
DenseNet [5]	High	91.48±0.52	6.97	2.25
MobileNetV2 [76]	High	90.70±0.37	2.24	0.27
NASNet [22]	NAS	NC*	4.25	0.51
DARTS [23]	NAS	87.28±0.25	3.96	0.48
MNASNet [28]	NAS	88.05±0.51	3.12	0.29
SGAS [27]	NAS	88.12±0.29	4.68	0.55
PDARTS [26]	NAS	90.42±0.78	4.18	0.49
RTRMM(BA-4)	NAS	91.38±0.49	0.82	0.25
RTRMM(ER-4)	NAS	**92.00±0.38**	**0.81**	**0.24**
RTRMM(WS-4)	NAS	**92.00±0.33**	0.82	**0.24**

* NC means "no convergence".

TABLE IV
EXPERIMENTAL RESULTS ON THE AID DATASET

Network	Time(h)	OA(%)	Params(M)	FLOPs(G)
GIST [62]	-	35.07±0.41	-	-
LDA (SIFT) [62]	-	56.52±0.41	-	-
IFK (CH) [62]	-	72.76±0.70	-	-
AlexNet [62]	-	89.53±0.31	40.35	5.02
VGGNet [62]	-	89.64±0.36	134.38	110.20
GoogLeNet [62]	-	86.39±0.55	5.63	10.80
ResNet [4]	-	91.16±0.37	23.57	29.99
DenseNet [5]	-	92.08±0.21	6.98	20.50
MobileNetV2 [76]	-	90.50±0.07	2.26	2.28
NASNet [22]	-	NC*	4.26	4.30
ALP [39]	-	87.00±0.75	3.75	-
DARTS [23]	10.7	89.08±0.19	3.97	3.99
MNASNet [28]	**7.1**	90.00±0.27	3.14	2.40
PDARTS [26]	9.8	90.70±0.49	4.20	4.09
SGAS [27]	12.0	91.56±0.42	4.69	4.57
RTRMM(WS-4)	9.5	93.81±0.24	0.82	**2.09**
RTRMM(ER-4)	10.4	93.88±0.25	**0.81**	2.12
RTRMM(BA-4)	11.1	**94.35±0.29**	0.83	2.15

* NC means "no convergence".

tion diversity is not abundant. Moreover, a successor node has only two edges with all predecessor nodes; that is the indegrees of successor nodes must and can only be two. These design patterns imply a stricter topology connection rule and contain a relatively strong human subjective factor. Excessive artificial design perhaps interferes with the performance of the model while a relatively flexible randomness approach may be more in line with the structure of biological neuron systems [77], [78], making the network construction engineering more effective. Furthermore, it is noteworthy that high-level feature-based methods and NAS methods are pretrained on ImageNet whilst the proposed three RTRMM models are just initialized through kaiming normal. As shown in [33], deep neural networks with pretrained weights can generally achieve better performance. And it is relatively difficult to train a deep model from scratch [79], comparing with the pretrained one. Hence, to sum up everything that has been stated so far, these three multiscale RTRMM models can extract critical parts of the features for remote-sensing images and they own a strong capability of feature expression for them. Besides, visualization heatmaps of the three RTRMM

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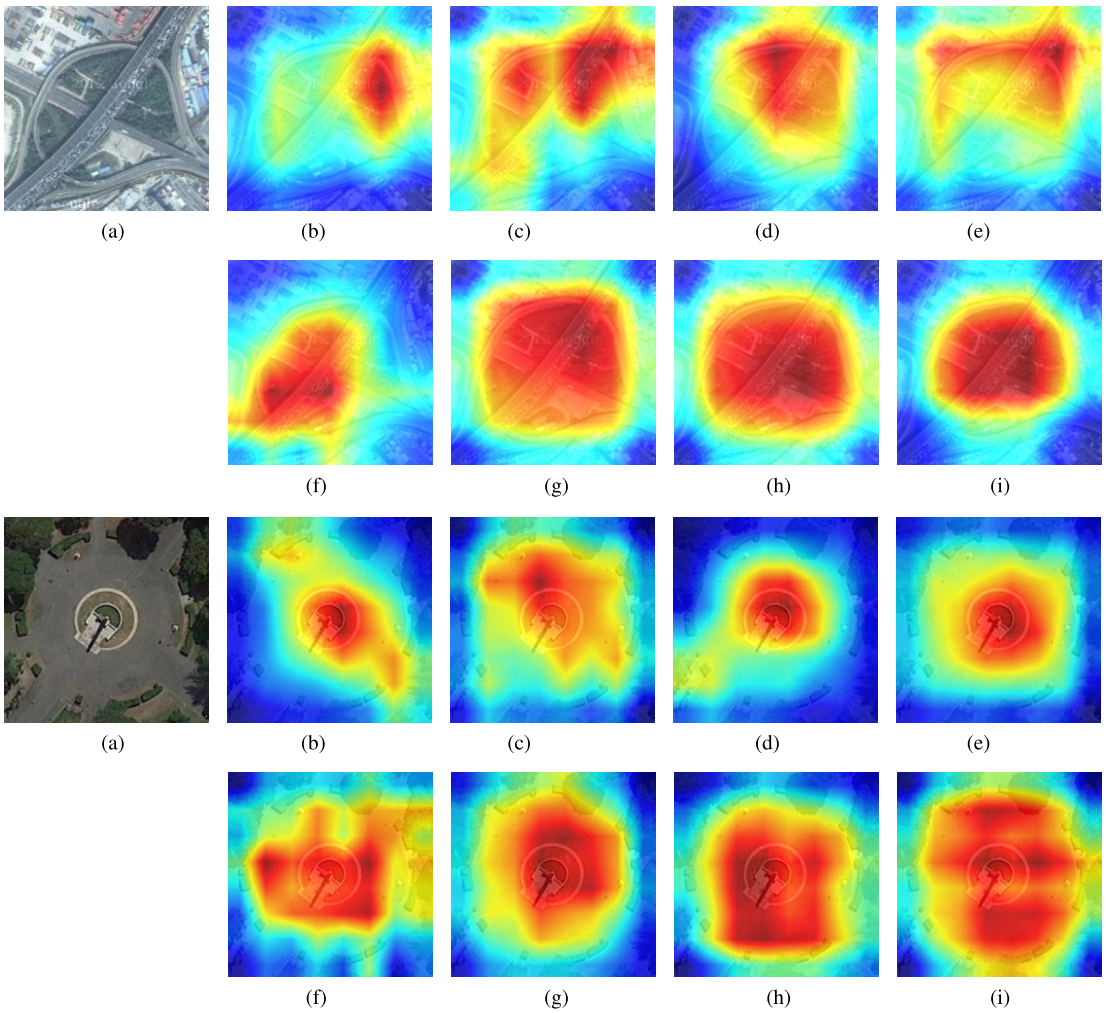


Fig. 10. Comparison of three RTRMM models and some baseline networks in visualization of heatmaps. (a) and (j) Raw remote sensing images. (b) and (k) Heatmaps of DenseNet. (c) and (l) Heatmaps of MobileNetV2. (d) and (m) Heatmaps of DARTS. (e) and (n) Heatmaps of MNASNet. (f) and (o) Heatmaps of PDARTS. (g) and (p) Heatmaps of ER-4. (h) and (q) Heatmaps of BA-4. (i) and (r) Heatmaps of WS-4.

TABLE V
EXPERIMENTAL RESULTS ON THE NWPU45 DATASET

Network	Time(h)	OA(%)	Params(M)	FLOPs(G)
AlexNet [63]	-	79.85±0.13	40.41	0.91
VGGNet [63]	-	79.79±0.15	134.44	20.21
GoogLeNet [63]	-	78.48±0.26	5.65	1.98
ResNet [4]	-	83.00±0.23	23.60	5.38
DenseNet [5]	-	83.22±0.14	7.00	3.77
MobileNetV2 [76]	-	82.69±0.06	2.28	0.41
NASNet [22]	-	67.48±0.10	4.28	0.77
SGAS [27]	6.6	75.87±0.75	4.70	0.81
DARTS [23]	4.9	79.18±0.15	3.98	0.71
MNASNet [28]	**3.3**	81.92±0.07	3.16	0.43
PDARTS [26]	5.0	82.14±0.53	4.21	0.73
RTRMM(ER-4)	5.5	85.66±0.41	**0.82**	**0.38**
RTRMM(WS-4)	5.0	86.19±0.77	**0.82**	**0.38**
RTRMM(BA-4)	6.1	**86.32±0.40**	0.83	0.39

models also explain this characteristic. As shown in Fig. 10, saliency regions which RTRMM models mainly concern about almost cover the key parts properly in different remote-sensing images while the behavior of some baseline networks are slightly deficient.

The comparison of running time for different models on two relatively large datasets, AID and NWPU45, are listed in Tables IV and V, respectively. The proposed approach achieves a better result, e.g., RTRMM(WS-4). It is just inferior to mobile neural architecture search net (MNASNet), but superior to or comparable with other methods. Relatively more parallel branches probably slow down the speed. While as for the computational complexity or lightweight aspect, the RTRMM models are quite competitive as well. The ER-4 architecture and the WS-4 architecture have a prominent performance. Due to the decomposition of convolution kernels, both them are much more compact than the other models, especially for some lightweight design models or NAS methods, and the computation cost is also significantly low. In terms of the quantity of network parameters, there are only fewer than 1 million to be optimized through gradient descent algorithm, only about a quarter of some NAS methods. In addition, FLOPs is also cost-effective, on the AID dataset, merely around 2.00 G for a very large-scale remote-sensing image with the size of 600×600 pixels. Compared with some NAS algorithms, there is a more obvious decline in terms of computational complexity for the proposed RTRMM models.

Besides these performance metrics as mentioned above, the corresponding confusion matrices computed on the best results are also given in Figs. 11–14. From these confusion matrices derived for the UCM, SIRI-WHU, and AID datasets,

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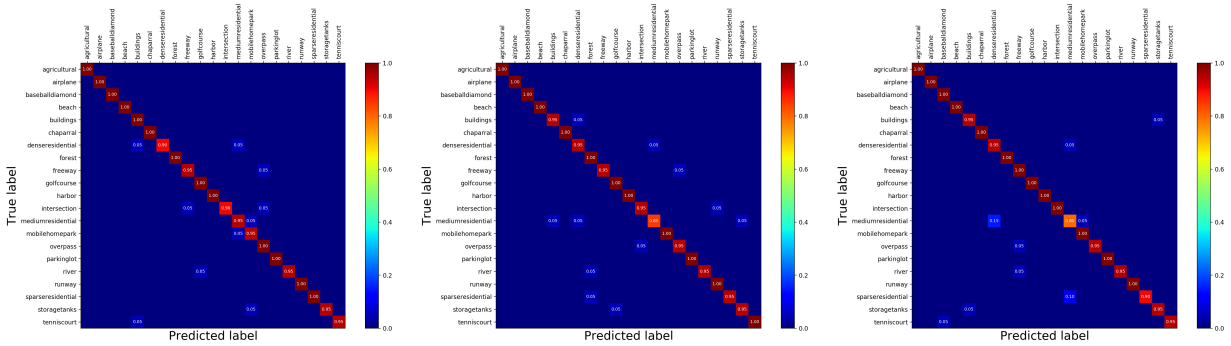


Fig. 11. Confusion matrices of three RTRMM models on the UCM dataset. (a) ER-4. (b) WS-4. (c) BA-4.

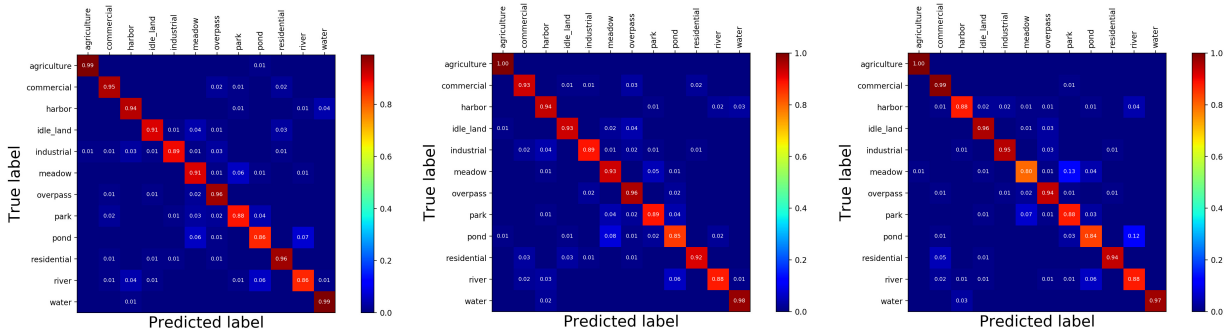


Fig. 12. Confusion matrices of three RTRMM models on the SIRI-WHU dataset. (a) ER-4. (b) WS-4. (c) BA-4.

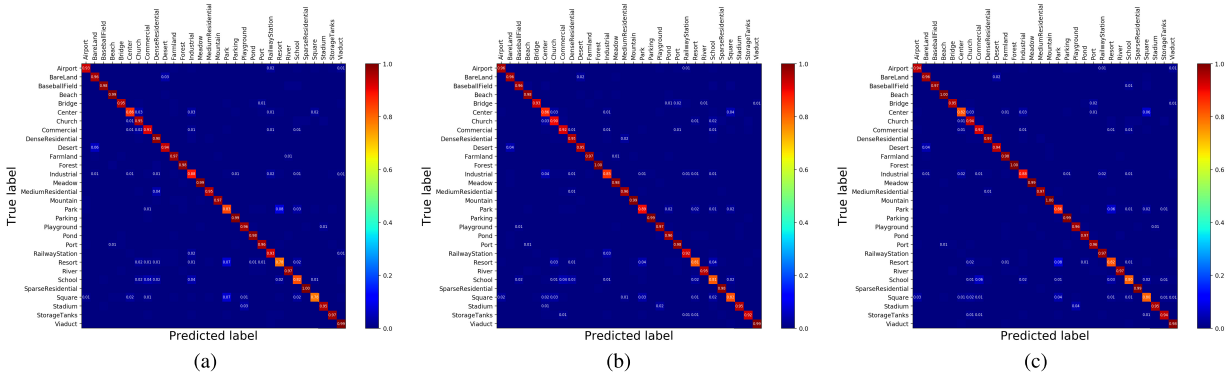


Fig. 13. Confusion matrices of three RTRMM models on the AID dataset. (a) ER-4. (b) WS-4. (c) BA-4.

we find that the class accuracies of most scene categories are higher than 0.90 or even close to 1.00 and only a small part of the scene categories have low-class accuracies. This means that different remote-sensing scenes are relatively easy to be distinguished from others for RTRMM models. Regarding the confusion matrices derived for the NWPU45 dataset, there is an obvious decline in classification behavior since the training data is insufficient; it only comprises 20% of the total dataset. Nevertheless, high classification accuracies can still be kept on most remote-sensing scenes. Most of the values of these three confusion matrices are still concentrated on the principal diagonal. It is also a satisfactory classification result with a relatively small training set. Hence, we can draw an important conclusion that the RTRMM models are still able to obtain the pivotal information of remote-sensing scenes even with very few samples.

E. Ablation Experiments

1) Multiscale and Lightweight Analysis: To verify the effectiveness of multiscale decomposition mapping, we conduct experiments on the effect of this part. Fig. 15 explains the performance comparison among single-scale models, some baseline models and multiscale models on four scene classification datasets. We pick the WS-4 architecture and we force the single scale of the WS-4 architecture to be 3, 5, 7, and 9,

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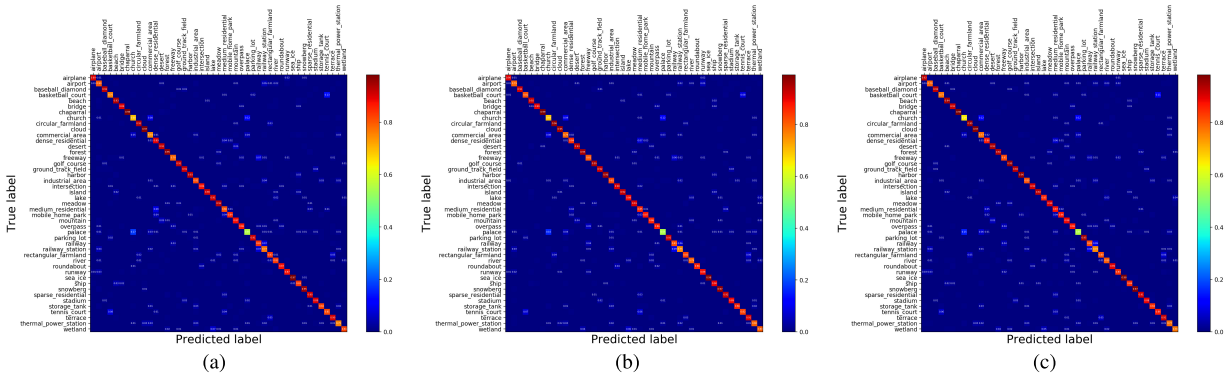


Fig. 14. Confusion matrices of three RTRMM models on the NWPU45 dataset. (a) ER-4. (b) WS-4. (c) BA-4.

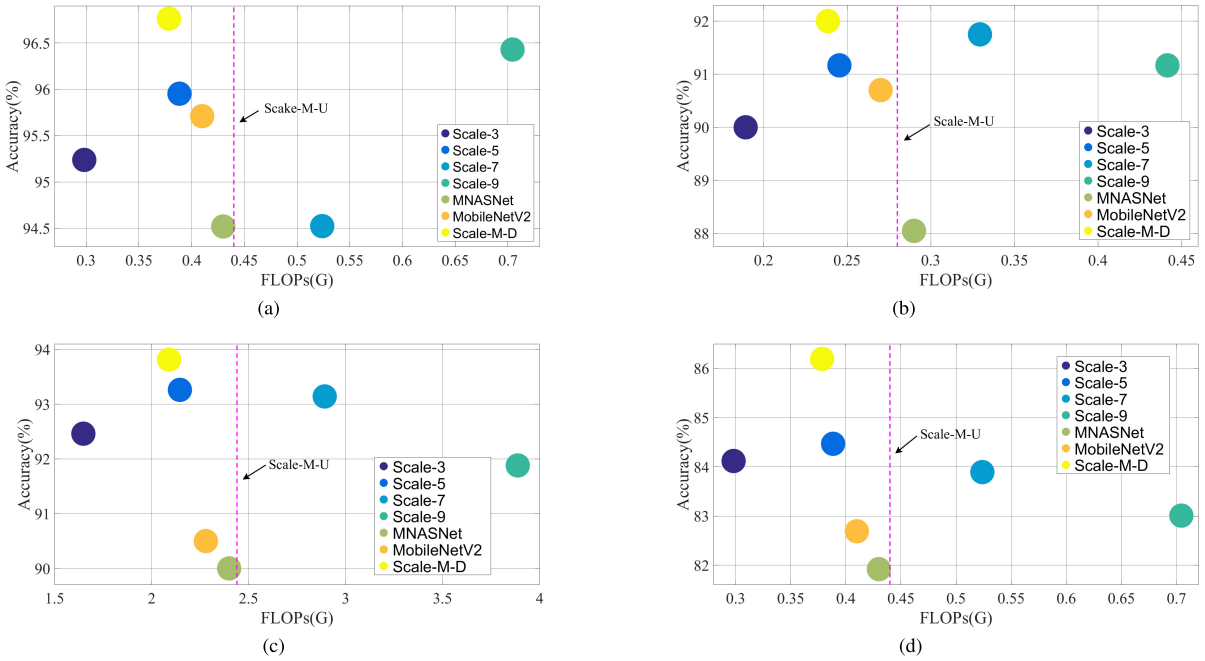


Fig. 15. Performance comparison of different scale models based on WS-4 and some baseline networks on four scene classification datasets. Solid circulars with different colors represent different models. Scale-M-D means multiscale models with decomposition and Scale-M-U means undecomposed models. Best view in color. (a) Comparison on the UCM dataset. (b) Comparison on the SIRI-WHU dataset. (c) Comparison on the AID dataset. (d) Comparison on the NWPU45 dataset.

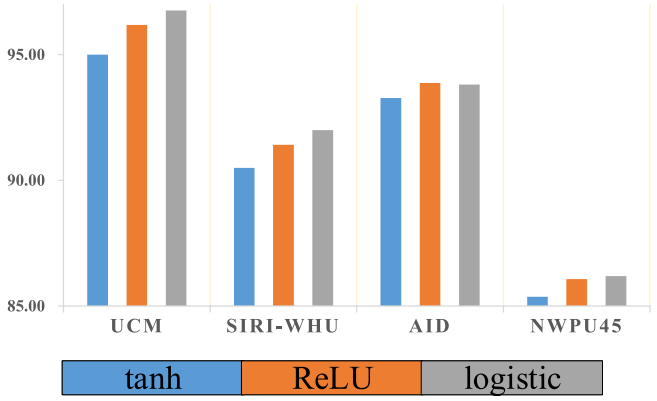


Fig. 16. Effect of different transfer functions, i.e., *tanh*, *ReLU*, and *logistic*, on model's performance on four datasets. Different colors represent different transfer functions. Best view in color.

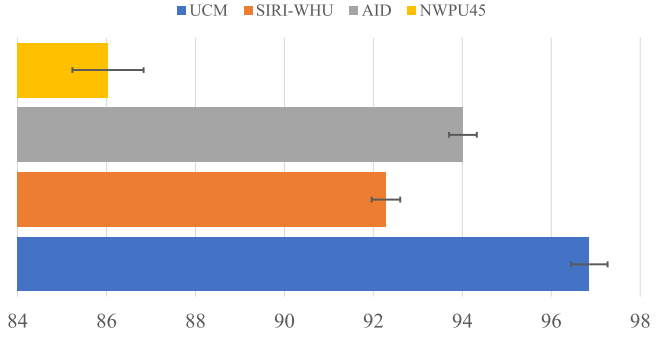


Fig. 17. Influence of different random seeds on model's performance on four datasets. Best view in color.

respectively. Then, we train these four models on UCM, SIRI-WHU, AID, and NWPU45 successively and evaluate their performance in terms of OA and FLOPs.

From the comparison, we can observe that multiscale model is able to make a better tradeoff between classification accuracy and computational complexity performance. Owing to the relatively small scale of the UCM dataset and the SIRI-WHU dataset, the improvement brought by the multiscale model is not obvious. Nevertheless, for AID and NWPU45 which are

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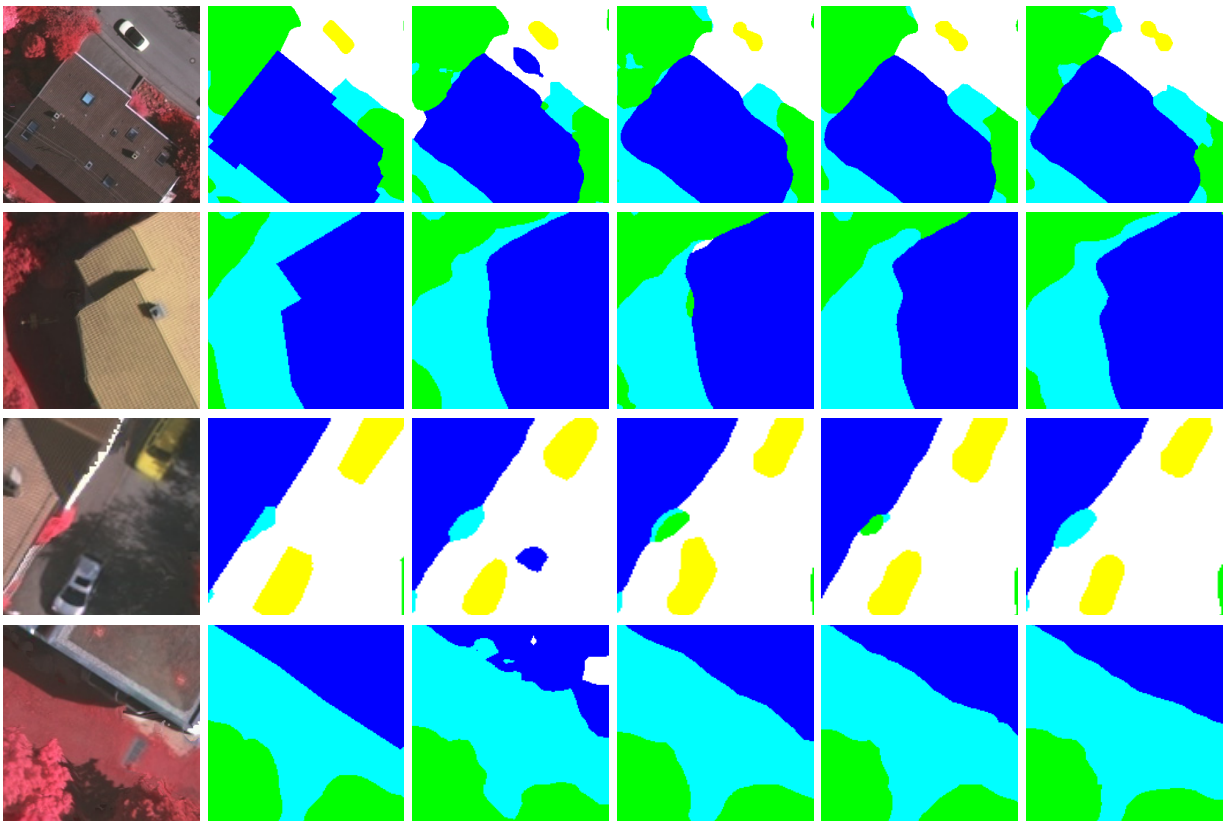


Fig. 18. Qualitative results on Vaihingen. Each column separately presents raw images, GT labels, results of original FCN and results of FCNs which take three proposed RTRMM models as the backbone. Legend–White: Impervious surfaces. Blue: Buildings. Cyan: Low vegetation. Green: Trees. Yellow: Cars. (a) Raw. (b) GT. (c) FCN. (d) ER-4. (e) BA-4. (f) WS-4.

both quite large-scale scene classification datasets, there exists notable improvements on FLOPs and accuracy separately. Therefore, the multiscale model remedies the unsatisfactory performance of single-scale models due to the relatively diverse scale information.

Furthermore, we can see that FLOPs of Scale-M is relatively high if we do not adopt the convolution kernel decomposition approach. It is noted that some lightweight networks are still very strong baselines. FLOPs of MobileNetV2 and MNASNet are just less than 0.50 G on the UCM, SIRI-WHU, and NWPU45 datasets. It is quite difficult to achieve this. Yet, after combining with decomposition strategy, this issue on multiscale models can be alleviated. FLOPs of Scale-M-D is on par with or even lower than for these excellent baseline networks.

2) Transfer Function Analysis: With regard to the choice of transfer function σ, we perform some comparison experiments on three commonly used functions (i.e., *tanh*, *ReLU*, and *logistic*) on UCM dataset, SIRI-WHU dataset, AID dataset, and NWPU45 dataset, respectively. As is stated above, we still pick the WS-4 network architecture. And, the transfer functions of the model are separately set to these three functions. The effect of different transfer functions on model's performance can be referenced in Fig. 16.

We can find in these experiments that for the larger and smaller remote-sensing images, it presents discrepant results for the effect of three selected transfer functions. For relatively large size images, such as AID dataset, the behaviors of three selected transfer functions are almost well-matched. There is little difference among them. Yet, concerning relatively small size images, for instance, UCM dataset, *ReLU* and *logistic* functions show noticeable gains compared with *tanh* function. While, in the comparison of the former two transfer functions, namely *ReLU* and *logistic*, *logistic* also achieves a slight performance improvement.

3) Random Seed Analysis: As for the influence of random seed s, we further have a discussion in this section. We set five random seeds to generate five different WS-4 networks and count their mean value and standard deviation on the aforementioned four remote-sensing scene classification datasets. Notice that other parameters, that is N, R, and P, for WS-4 are all consistent with previous experiments. The results are shown in Fig. 17.

It can be seen that the fluctuation near the mean value is relatively small on SIRI-WHU dataset and AID dataset. The performances of different models are very close to each other. While, owing to relatively insufficient training samples for NWPU45 dataset, the stability of the model is greatly disturbed. The standard deviation of the performance is relatively large, but these models are still very competitive and are able to keep a relatively high OA value as well.

F. Semantic Segmentation Experiments

1) Quantitative Results: Table VI shows comparison results achieved for the ISPRS Vaihingen dataset. It can be noticed

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TABLE VI
QUANTITATIVE SEGMENTATION RESULTS ON THE ISPRS VAIHINGEN DATASET USING THE FCN FRAMEWORK

Network	Imp.surf.	Build.	Low veg.	Tree	Car	mF1	mIoU	OA	Params(M)
FCN (original) [67]	88.67	92.83	76.32	86.67	74.21	83.74	72.69	86.51	134.30
FCN (ER(32, 0.2)-4)	88.32	92.26	77.67	86.87	73.63	83.75	72.65	86.51	**1.55**
FCN (BA(32, 5)-4)	88.56	92.45	**77.87**	**86.97**	74.06	83.98	72.99	86.72	1.57
FCN (WS(32, 4, 0.75)-4)	**89.22**	**93.03**	77.14	86.69	**75.11**	**84.24**	**73.39**	**86.84**	1.56

that there is an improvement on the performance of FCNs with RTRMM models as their backbone in the case of very few parameters, especially for the WS-4 architecture. The quantity of parameters in FCN(WS-4) is only 1.16% of the one in the original FCN. The number of parameters is reduced obviously, which makes the obtained network become more compact and lightweight. For the performance regarding small objects such as given in the car category, the F_1-score of it evaluated on FCN(WS-4) has about one point increment compared to the original FCN.

2) Qualitative Performance: Fig. 18 presents several visualizations regarding the performance on test samples. From these results, we can see that the boundary information are more clear in the modified FCNs. In the results on the first, third, and fourth lines, there are serious segmentation errors while for other modified FCNs, there is no such performance. And in the second line, the edges of this blue object are becoming sharp.

Consequently, from the analysis above, we can come to a conclusion that the features learned by three proposed RTRMM models can also be transferred to the remote-sensing semantic segmentation task. And these features are more superior and generic, which makes the segmentation networks perform better at the edges and details.

V. CONCLUSION

In this article, we first review some applications of NAS in natural scene image and remote-sensing field. Owing to the complexity of remote-sensing image, such as rotation effect, rich background information, scene similarity, etc., the performance improvement is relatively limited in remote-sensing field for some NAS approaches, though they have made remarkable achievements in natural scene image. Besides, there is also a problem of high-computational complexity for some NAS methods applied to remote sensing. Therefore, we propose a RTRMM method to generate a more lightweight neural network architecture with a capability of multiscale feature extraction.

Specifically, our method designs a RT generator that leverages three classical random graphs, i.e., ER, WS, BA, to automatically construct topological relationships of neural networks. To satisfy the normal data flow, we define three types of nodes and some connection strategies. Meanwhile, a RMM sampler is also proposed. We design a multiscale search space which consists of decomposition convolution units. The largest scale in the search space is nine and the maximum scale will decrease with the deepening of the network, for the size of feature maps becomes smaller gradually. Through the mathematical analysis about model complexity, we find that decomposition convolution units can make the generated model more compact and lightweight. For fairness and objectivity, we select the final mappings through random sampling with replacement. Ultimately, the three models **ER(32, 0.2)-4**, **WS(32, 4, 0.75)-4**, and **BA(32, 5)-4** are obtained.

Experiments on four widely used remote-sensing scene classification datasets i.e., UCM, SIRI-WHU, AID, NWPU45, confirm the feature extraction capability of the proposed RTRMM models. It actively reveals that random design is a very competitive approach. And from the comparison of the number of parameters and FLOPs, we can see that RTRMM models are very lightweight as well. Besides, we find that our approach can achieve a better tradeoff between classification performance and lightweight performance. Furthermore, for FCNs whose backbone is replaced by RTRMM models, there exists an improvement in semantic segmentation for the ISPRS Vaihingen dataset in the case of very few parameters. And edges and details are also more clear in qualitative results. These segmentation results verify the high feature-transfer capability of RTRMM models. For future work, we will continue to concentrate on automated and lightweight network design, and attempt to apply our method to other remote-sensing visual tasks, such as fine-grained classification.

ACKNOWLEDGMENT

The authors deeply appreciate the effort of the anonymous reviewers and would like to thank them for their valuable comments.

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